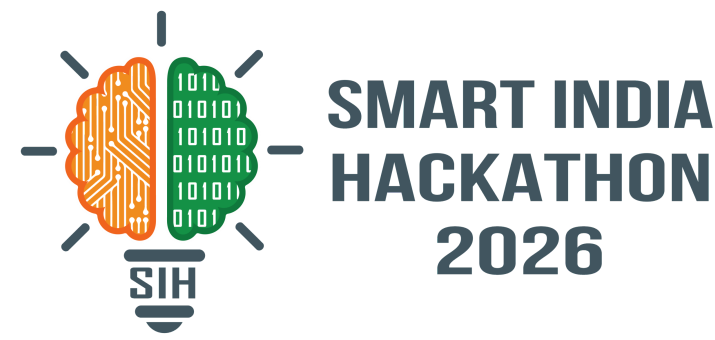
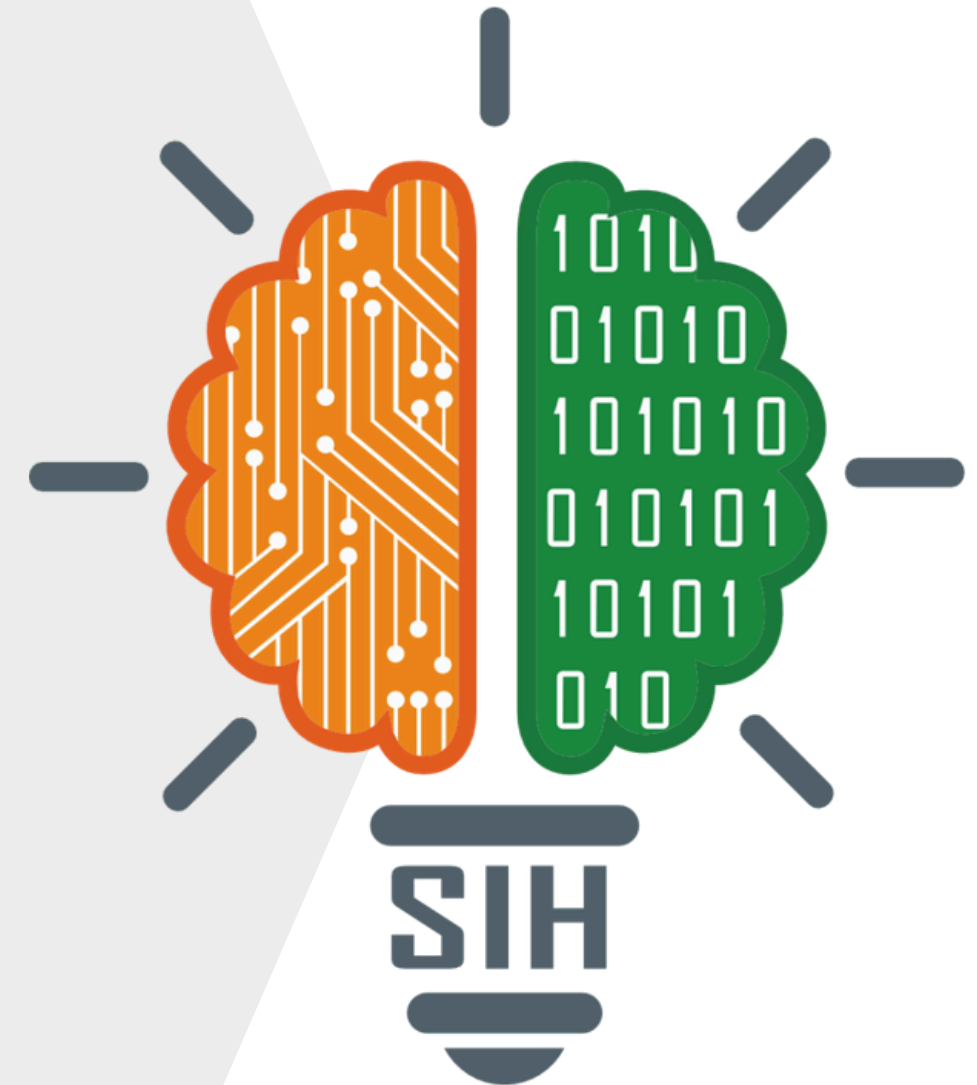


FAB 6

SMART INDIA HACKATHON 2026



- **Problem Statement ID : 26039**
- **Name: Nischall S Haritas, Rishit Negi, Jeffrey Simon Chris,
D Dharnesh, Theertha Santhosh, Navya S N**
- **Problem Statement Title: AI-Powered Underground Mine Safety,
Monitoring and Rescue System**
- **Theme: Smart Automation**
- **PS Category: Hardware**
- **Team ID: SIH_CU_010**
- **Team Name: FAB6**





Problem:

1. Toxic and hazardous gases
2. Mine structure and obstacles
3. Darkness and poor visibility
4. No reliable GPS underground
5. Communication loss
6. Trapped or missing workers



CORE INNOVATION

AI Hazard-
Mapping Engine

PRIMARY FUNCTIONS

Gas & Air-Quality Sensing |
Autonomous Navigation

SAFETY & COMPLIANCE

Collapse Alert | O₂ Alarm | SOS Beacon

MONITORING & SUPPORT

Thermal | Humidity | Obstacle Map | Live Video & Audio

RISK

Toxic Gas
Exposure

Structural
Collapse

Delayed Rescue
Response

Communication loss/
No GPS

SOLUTION

Real-Time Gas
Sensing & Alerts

MPU6050
vibration/shock
detection

Autonomous drone
first reconnaissance

Tethered link (current
prototype) + LoRa relay
nodes (planned wireless
extension)



SENSE

COMMUNICATE

ANALYZE

ALERT

ASSIST RESCUE



TECHNICAL APPROACH



TECHNOLOGIES USED



Flight Platform

F450 quadcopter frame + flight controller



Microcontroller

ESP32 — dual-core, Wi-Fi / BLE



Sensor Suite

MQ-2/MQ-4/MQ-7 gas, DHT22 , MPU6050, HC-SR04, MLX90614 + scanning servo.



Communication

Tethered link (demo reliability) + LoRa relay nodes



Software / AI

Edge Impulse TinyML + Gemini via Google AI Studio (live dashboard)

FEASIBILITY ANALYSIS

- **Infrastructure Reuse:** Off-the-shelf F450 frame & ESP32 keep hardware cost low.
- **Modular Sensors:** Sensor pods added in phases as budget/testing allow.
- **Zone-Based Mock Mine:** a man-made test structure with dedicated gas, high-temperature, and thermal/trapped-worker zones — validates each sensor against a controlled hazard without needing access to a real mine.
- **Lightweight Payload:** Keeps flight time and manoeuvrability practical underground.

VIABILITY

- **Proven Precedent:** Drone-based inspection already used in oil & gas and civil-construction safety audits.
- **Reduces Human Bias:** Automated sensing standardises hazard calls vs. manual judgement.
- **Auditable Records:** Real-time telemetry logs create a traceable digital safety record.
- **Regulator-Friendly:** Aligns with DGMS push for tech-enabled mine safety compliance.

BUSINESS POTENTIAL

- **Ready Budgets:** Coal India, SAIL & private operators already fund statutory safety compliance
- **Flexible Model:** Drone-as-a-Service or one-time CAPEX sale to mine operators.
- **Recurring Revenue:** Subscription analytics dashboard across multi-site fleets.
- **Export Potential:** Adaptable to tunnelling, metro & underground infra safety beyond mining.

FAB 6

IMPACT AND BENEFITS

FROM BLIND RESCUE TO INTELLIGENT SITUATIONAL AWARENESS



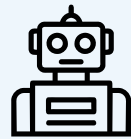
PROTECT RESOURCES

Remote reconnaissance helps assess hazardous areas before human entry.



FASTER DECISION-MAKING

Real-time environmental and visual information helps rescue teams assess conditions quickly.



AI-ASSISTED DETECTION

Edge Impulse flags structural anomalies on-device; Gemini synthesizes sensor data into hazard alerts



IMPROVED VISIBILITY

RGB (FPV) + scanning thermal (MLX90614)



REAL-TIME MONITORING

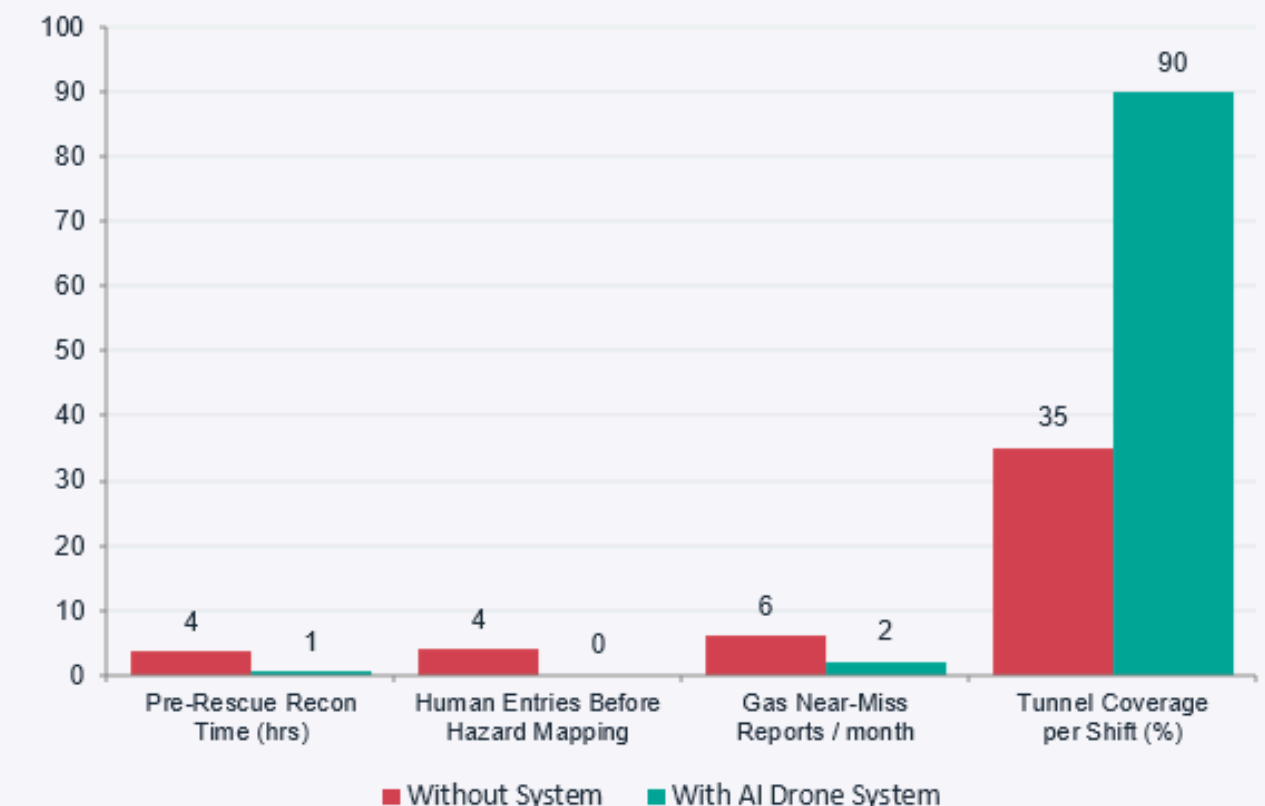
Sensor readings, video and alerts are brought together at the control station.



SCALABLE SYSTEM

Modular hardware allows additional sensors and AI capabilities to be integrated later.

PROJECTED IMPACT (illustrative estimates)



DGMS Safety Statistics

data.gov.in-DGMS

Gas & Air Quality Sensor Design

espressif.com (ESP)
32 docs

TinyML Structural Anomaly Detection

edgeimpulse.com

LoRa Mesh GPS Denied Zone

semtech.com/lora

Drone-Based Mine Monitoring Research

ieeexplore.ieee.org

Jharkhand Mining & Safety Guidelines

jharkhand.gov.in

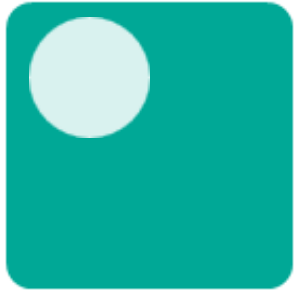
SYSTEM CONCEPT — CONTROL ROOM DASHBOARD (mockup)



Node A- Gas Level

CH₄: 0.4% | CO:12ppm

SAFE



Node B- Humidity

78% RH | 31°C

NORMAL



Node C- Obstacle Dist

1.8 m ahead

CAUTION



Node D- SOS status

No active alerts

STANDBY